Graph Relation Aggregated Spectral Perception Network for Hyperspectral Classification

Jiaojiao Li, *Senior Member, IEEE*, Xinyi Yan, Haitao Xu, *Member, IEEE*, Chengyuan Wang, Rui Song, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*—Cross-domain few-shot learning hyperspectral image classification (CDFSL HSIC) presents significant challenges, primarily due to spectral variability across domains and the scarcity of labeled target samples. To tackle these issues, we propose a graph relation aggregated spectral perception network (GRASP-Net), which progressively models spectral and structural information from the sample level to the category level. First, an enhanced spectral perception (ESP) module is developed to refine sample-level features by enhancing discriminative spectral cues while suppressing redundancy. On this basis, a prototype-guided graph construction (PGC) module establishes a sample–prototype bipartite graph, integrating geometric neighborhood relations with nongeometric priors to generate compact and robust category-aware embeddings. Furthermore, a gradient-aware metric function interprets gradients as directional vectors to adaptively refine sample–prototype relations, thereby overcoming the limitations of conventional scalar distance metrics. Extensive experiments on multiple cross-domain benchmarks demonstrate that GRASP-Net achieves superior accuracy and robustness compared with state-of-the-art methods, particularly in scenarios with limited labeled samples and large domain shifts.**

***Index Terms*—Cross-domain few-shot learning (CDFSL), hyperspectral image classification (HSIC), prototype network.**

I. Introduction

HYPERSPECTRAL image (HSI) constitutes a 3-D cube containing rich spectral information, which captures detailed spectral signatures for each pixel and finds strong correlations with geophysical analysis. Owing to its discriminative spectral characteristics, the HSI has exhibited considerable potential across numerous domains [1], such as remote sensing image analysis [2], environmental monitoring [3], mineral exploration [4], and so on. To fully harness the rich spectral information embedded in HSIs, HSI classification (HSIC) serves as a fundamental processing technique. It assigns semantic labels to each pixel by analyzing the spectral properties, thereby converting raw data into meaningful interpretations that support accurate and fine-grained land-cover analysis.

Traditional HSIC methods, mainly based on classic classification algorithms [5], [6], [7], [8] or feature engineering [9], [10] struggle to model complex spatial–spectral correlations and handle the high dimensionality of HSIs. Convolutional neural networks (CNNs) have achieved strong performance in HSIC [11] due to the unique convolution operation and hierarchical structure. Early works used 1D-CNNs and 2D-CNNs for spectral or spatial modeling, while recent studies adopt 3D-CNNs [12] to jointly extract spatial–spectral features, with subsequent works exploring adaptive feature enhancement mechanisms to strengthen discriminative representations [13]. Beyond convolution-based approaches, alternative representation paradigms have also been explored in remote sensing [14]. However, CNNs typically require large labeled datasets to generalize well. In practice, HSI data and pixel-level annotations are scarce due to high acquisition and labeling costs, limiting large-scale supervised learning. Consequently, few-shot learning (FSL) [15] has gained attention for its ability to learn transferable representations from limited labeled samples and adapt to new classes with minimal supervision. Existing FSL methods are typically categorized into model-based, optimization-based [16], and metric-based methods. Among them, metric-based methods [17], [18] have gained significant popularity, as they learn an embedding space with a suitable metric function [19] to reduce intraclass distance and increase interclass separability. A representative example is prototypical network [20], which simplifies distance computation and class assignment by estimating prototype vectors as cluster centers of various classes [21]. For HSIC, DFSL [22] designs a 3-D residual CNN (3D-ResNet) to better extract spatial–spectral joint features without preprocessing. Furthermore, S3Net [23] is proposed to balance the number of positive and negative samples, thus expanding the training dataset.

In real-world scenarios, HSIs are often acquired by different sensors across diverse geographical regions. Variations

Received 9 December 2025; revised 5 March 2026; accepted 23 March 2026. Date of publication 26 March 2026; date of current version 3 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62371359, in part by Shaanxi Provincial Science and Technology Department under Grant 2024GX-ZDCYL-02-09, in part by Xi'an Science and Technology Bureau under Grant 2024JHCLYB-0106, in part by the "Scientist + Engineer" team of the Qin Chuang Yuan in Shaanxi Province, in part by Xi'an Science and Technology Program, and in part by the "Leading the Charge" Initiative for the Industrialization of Core Technologies in Key Industrial Chains in Shaanxi Province. *(Corresponding authors: Haitao Xu; Chengyuan Wang.)*

Jiaojiao Li, Xinyi Yan, and Rui Song are with the State Key Laboratory of Integrated Service Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: jjli@xidian.edu.cn; yxymian@stu.xidian.edu.cn; rsong@xidian.edu.cn).

Haitao Xu is with the National Space Science Center, Chinese Academy of Sciences, Beijing 100864, China (e-mail: xuhaitao@nssc.ac.cn).

Chengyuan Wang is with the Information and Communication College, National University of Defense Technology, Changsha 410073, China (e-mail: wangchengyuan@nudt.edu.cn).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2026.3677864

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

in environmental conditions, vegetation coverage, and terrain morphology lead to substantial domain discrepancies in both spatial and spectral characteristics, as well as semantic misalignment across domains. These issues significantly exacerbate the difficulty of FSL HSIC. To tackle this issue, the topic of cross-domain FSL HSIC (CDFSL HSIC) [24] is introduced. In this setting, the domain with sufficient labeled samples is referred to as the source domain (SD), while the domain with very limited annotations is termed the target domain (TD). The substantial domain gap, coupled with the scarcity of labeled data in the TD, poses critical challenges for CDFSL-HSIC. In light of this challenge, methods including transfer learning and domain adaptation [25] have been proposed, leveraging SD knowledge to improve TD performance. For instance, deep cross-domain FSL (DCFSL) [26] aligns SD and TD samples in a shared embedding space to enhance generalization. Inspired by the metric-based FSL concept, Xi et al. [27] proposed an FSL method based on class covariance metrics (CMFSL), which utilizes a spectral prior refinement module and employs class covariance metrics to better domain discrepancies. Recent works further highlight that CNN-based models insufficiently exploit the graph-structured information in HSIs. To address this, Zhang et al. [28] introduced Gia-CFSL to align cross-domain graph information, while Ding et al. [29] proposed GLGAT-CFSL by integrating convolutional operators with graph neural networks to achieve more comprehensive domain alignment.

Current HSIC methods mainly focus on modeling local geometric structures but often overlook global relational information, including high-level topological patterns and interclass semantic dependencies. These global cues are essential for maintaining coherent feature organization. The topological patterns preserve the overall geometry of the class clusters, while semantic dependencies regulate their separation and interactions. Without explicitly modeling such relations, local features become inconsistently organized under domain shift, leading to ambiguous boundaries and weakened cross-domain transferability. To overcome these limitations, we propose the graph relation aggregated spectral perception network (GRASP-Net) for CDFSL HSIC. The framework constructs a hierarchical pipeline that progresses from sample-level spectral perception to category-level graph aggregation. At the sample level, the enhanced spectral perception (ESP) module performs dynamic spectral reorganization with a dual-branch design (kernel-attentive and kernel-scattered), enabling the model to capture fine-grained spectral variations and enhance the discriminability of individual samples. Building on these refined representations, the prototype-guided graph construction (PGC) module operates at the category level, where a sample–prototype bipartite graph integrates geometric neighborhood relations with nongeometric category priors, yielding compact and robust category-aware features. Furthermore, since conventional metric functions rely solely on scalar distances and thus lack directional information to model sample–prototype relations, we introduce a gradient-aware metric function. By treating gradients as force-like vectors, it provides finer relation refinement, improves the separability of confusing samples, and enhances the reliability of classification in challenging cases.

The main contributions of this article are as follows.

1) We propose the GRASP-Net for CDFSL HSIC, establishing a hierarchical pipeline that progresses from sample-level spectral perception to category-level graph aggregation and further incorporates gradient-aware refinement.
2) The ESP module is designed to perform dynamic spectral reorganization through a dual-branch strategy (kernel-attentive and kernel-scattered), allowing subtle spectral variations to be amplified and the discriminability of individual samples to be improved.
3) The PGC module is developed to construct a sample–prototype bipartite graph that integrates geometric neighborhood relations with nongeometric category priors, resulting in compact and robust category-aware features.
4) A gradient-aware metric function is introduced to address the limitations of conventional scalar distance-based metrics. By interpreting gradients as force-like vectors, it adaptively refines sample–prototype relations, enhances the separability of challenging samples, and overall classification reliability.

The remaining sections of this article are organized as follows. Section II delves into the related work on CDFSL HSIC. Section III provides a detailed elaboration on the network flowchart of GRASP-Net, elucidating how it leverages distinct receptive field branches to enhance the perception of spectral information in HSIs, and how it integrates both geometric and nongeometric information to guide HSIC. Section IV presents the experimental results and analysis, validating the efficacy of the proposed algorithm. Lastly, Section V concludes by summarizing the overall principle of the algorithm and outlining potential avenues for future improvements.

II. Related Works

A. Cross-Domain FSL

FSL, inherently a form of meta-learning, poses the central challenge of conducting effective learning under extreme data scarcity and achieving robust performance in deep learning models with minimal data. It employs episodic learning strategies for meta-learning, leveraging prior knowledge to rapidly generalize from limited supervised experience to new tasks. During the meta-training phase, the training dataset is partitioned into distinct meta tasks, each encompassing various categories. By sequentially training on these meta tasks, the model acquires the capability to classify unseen novel categories during the meta-testing phase. Specifically, for the N-way K-shot classification task, N classes are randomly selected from the dataset, and within each selected class, K training samples are randomly selected to constitute the support set for the meta-task, where K is typically a small number (e.g., 1 or 5). Subsequently, a fixed number of samples is randomly drawn from the remaining data of each class to form the query set serving to evaluate the model's learning efficacy.

However, there typically exist two domains in classification tasks [30], [31], the SD, which encompasses abundant labeled

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

data, and the TD, characterized by scarce labeled data and a preponderance of unlabeled data. A significant challenge in CDFSL arises when the categories in the SD and TD do not overlap, the labeled samples in the TD are scarce, and there exists a pronounced domain shift between the two domains. Accurately classifying data in the TD under these conditions poses a formidable task. Generally, two approaches are employed to address the issue of domain shift: fine-tuning and transfer learning. Fine-tuning involves training a network on a large dataset and subsequently refining it with a small dataset [32], allowing parameters to continue evolving. On the other hand, transfer learning fixes a portion of the parameters while updating the rest. A prevalent method within transfer learning is domain adaptation [33]. Zheng et al. [34] proposed a novel two-stage adaptation network (TSAN) algorithm that employs adversarial learning to make the classifier unable to distinguish the SD from the mixed multi-TDs. Ghifary et al. [35] first conducted supervised training on the SD, followed by reconstruction on the TD, and finally applied domain adaptation techniques.

B. Prototypical Network

The prototypical network is a representative approach in FSL [36]. It constructs an embedding space in which each class is characterized by a prototype vector, defined as the centroid of its support samples. During training, the network maps samples into the embedding space and optimizes a metric function so that the learned prototypes approximate the true class centers. In evaluation, query samples are classified by computing their distance to all class prototypes and assigning them to the nearest one following the nearest-neighbor rule. The prototype vector is formulated as follows:

$$C_1 = \frac{1}{\mathrm{n}_l} \sum_{(x_l^i, y_i) \in S_l} f_\theta \left(x_l^i \right) \tag{1}$$

where x_l^i represents the ith sample belonging to the support set S_l with label l, y_i is the true label of the sample x_l^i, $f_\theta(\bullet)$ denotes the feature extractor that maps data into features in the metric space, θ is the corresponding parameter, and C_1 is the prototype vector. Generally, the prototype vector is obtained by averaging the features of the same class within the support set.

C. Label Propagation

The label propagation algorithm fundamentally leverages the connectivity among nodes in a graph, iteratively propagating labels from labeled nodes to unlabeled ones for prediction. It suits large-scale graph data well due to its efficiency and intuitive logic, and has been widely adopted in practical tasks. For example, Yang et al. [37] addressed the high cost of pixel labeling and underutilized unlabeled data by integrating label propagation and contrastive regularization.

At the beginning of the algorithm, all nodes in the graph are initialized. For nodes with known labels, their original labels are preserved, while for nodes with unknown labels, a unique temporary label is assigned. In each iteration, each node updates its own label based on the label status of its neighboring nodes. The update rule typically involves selecting the most frequently occurring label among its neighbors as its new label. If there are multiple labels with the highest frequency, one of them can be chosen arbitrarily as the new label.

Label propagation essentially utilizes sample data to explore the graph structural relationships within the dataset, gradually transmitting the information of unlabeled data along the edges of the graph. By updating the node labels and the weights between nodes, label propagation can transform dispersed sample points in the metric space into a unified graph structure, enabling the model to learn nongeometric patterns in addition to geometric information.

III. Proposed Method

The overall architecture of the proposed GRASP-Net is illustrated in Fig. 1. Motivated by the need to enhance spectral perception, exploit category-aware structural cues, and refine metric-based decision boundaries in CDFSL HSIC, the framework is designed with three complementary components that correspond to the main contributions of this work. First, the ESP module employs a dual-branch strategy, named kernel-attentive and kernel-scattered branch to dynamically reorganize spectral information and extract discriminative sample-level representations. Second, the PGC module constructs a sample–prototype bipartite graph, enabling the integration of geometric neighborhood relations with nongeometric category priors for compact and robust category-aware embedding. Finally, a gradient-aware metric function is introduced to refine sample–prototype interactions from an optimization viewpoint, where gradient directions act as force-like cues that enhance separability and stabilize metric-based classification. These components form a hierarchical pipeline that transitions from fine-grained spectral perception to structured graph aggregation and gradient-level refinement.

A. Source and Target FSL

This section primarily presents the relevant settings for the cross-domain FSL task in this article. First, for dataset settings, the cross-domain scenario involves two domains: the SD with abundant labeled data, denoted as the SD dataset D_S, and the TD with limited labeled data, denoted as the TD dataset D_T. Second, for the few-shot task settings, we employ the episodic training paradigm. In each episode, a support set D_s^l and a query set D_q^l are constructed, where l indicates the origin of the data (either the source S or TD T). For sample settings, both the SD dataset and the TD dataset sample N classes for the corresponding episode, forming the support set and the query set, respectively. Given the scarcity of labeled samples in the TD, m samples are selected from each class in the TD and expanded to 200 samples per class via data augmentation techniques. In the TD episode, K samples are first selected from each class to form the support set, followed by Q samples selected from the remaining samples per class to serve as the query set. To summarize, taking the SD episode as an example, $D_s^S = \{(x_i, y_i)\}_{i=1}^{N \times K}$ represents the support set in a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

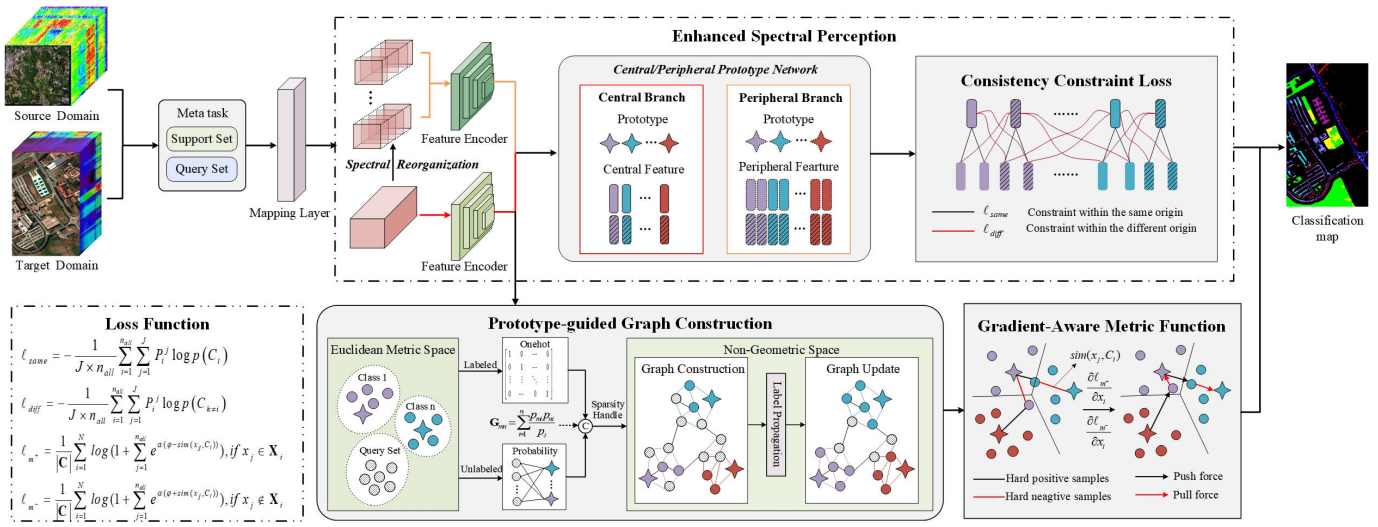


Fig. 1. Flow of the proposed GRASP-Net for CDFSL HSIC. The framework constructs a hierarchical pipeline that first enhances sample-level spectral perception through the ESP module, then aggregates category-level features with the PGC module, and finally refines sample–prototype relations via the gradient-aware metric function.

single episode, while $D_q^S = \{(x_k, y_k)\}_{k=1}^{N\times Q}$ represents the query set.

B. Enhanced Spectral Perception

HSIs comprise hundreds of continuous spectral bands that encode fine-grained physical and chemical characteristics of ground objects. Although such spectral details are crucial for class discrimination, existing methods often model spectra using fixed sequential structures, which introduce smoothing effects and local dependencies. These limitations suppress narrow-band variations, high-frequency components, and long-range correlations, leading to weakened discriminative representations and confusion among spectrally similar classes. To overcome these issues, we propose an ESP module that dynamically reorganizes spectral dimensions and adopts a dual-branch design to extract complementary spectral representations, thereby strengthening sample-level spectral perception.

1) Kernel-Attentive Branch: This branch processes the input following the original spectral order, aiming to capture dominant discriminative patterns across the full spectral range. Since the key class-specific differences are often concentrated in a limited set of salient bands, this branch can form stable intraclass clusters in the high-dimensional spectral space, leading to compact and discriminative embeddings. By modeling global correlations along the continuous spectral sequence, it effectively preserves fine variations and enhances separability among spectrally similar classes.

2) Kernel-Scattered Branch: Complementary to the kernel-attentive branch, the kernel-scattered branch increases the model's sensitivity to subtle spectral variations that are dispersed across distant bands. By exposing the network to diverse spectral arrangements and breaking sequential locality, it captures fine-grained cues that are often muted in standard sequential modeling. This complementary perspective enhances the separability of spectrally similar categories, mitigates boundary confusion, and improves the robustness and generalization of sample-level spectral representations.

To achieve this, the branch employs a perturbation-based strategy that generates J variant views by randomly sampling and recombining spectral bands from the input of the kernel-attentive branch. Each fragmented view retains only a subset of bands and reorganizes them in diverse ways before being fed into the kernel-scattered branch. This process disrupts local dependencies and continuously exposes the network to heterogeneous spectral modalities, compelling it to capture subtle and dispersed spectral variations. Ultimately, the kernel-scattered branch outputs a probability distribution derived from prototype-based metric information, which complements the global aggregation of the kernel-attentive branch and enhances overall discriminability.

3) Consistency Constraint: The dual-branch architecture captures complementary spectral perspectives. To leverage the discriminative cues from both branches, a consistency constraint is introduced to align their category prediction distributions. Specifically, cross-entropy is used to measure the consistency between the kernel-attentive and kernel-scattered branches for the same sample. Each branch independently produces prototype-based predictions, while their interaction is enforced at the prediction level through the consistency loss rather than explicit feature fusion. Moreover, as the kernel-scattered branch operates on perturbed spectral fragments and encodes complementary information, the consistency constraint implicitly regularizes the kernel-attentive branch to learn more reliable sample-level representations, thereby preserving branch complementarity and enhancing overall discriminability

$$\ell_{same} = -\frac{1}{J \times n_{all}} \sum_{i=1}^{n_{all}} \sum_{j=1}^{J} P_i^j \log p\left(C_i\right) \tag{2}$$

$$\ell_{diff} = -\frac{1}{J \times n_{all}} \sum_{i=1}^{n_{all}} \sum_{j=1}^{J} P_i^j \log p\left(C_{k\neq i}\right). \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

In ℓ_{same}, the cross-entropy loss measures the consistency between the class prediction scores C_i of sample i and P_i^j of its J variant views from the same origin. In ℓ_{diff}, the loss compares the consistency between the scores $C_{k\neq i}$ from another original sample k (where k and i belong to the same class) and scores P_i^j of the J variant views from sample i.

C. Prototype-Guided Graph Construction

In the CDFSL HSIC task, the sample-level ESP module optimizes individual feature representations through a dual-branch dynamic spectral reorganization strategy. However, accurate recognition further requires modeling category-level relationships. Conventional prototype-based metric approaches mainly emphasize geometric neighborhood relationships while overlooking nongeometric category priors inherent in HSIs, such as cross-domain interclass semantic correlations, distributional commonalities of samples in the prototype space, and implicit category association patterns that remain relatively stable across domains. This limitation weakens the global organization of category-level features and makes them vulnerable to fragmentation under domain shift.

To address this issue, we propose the PGC module, which focuses on category-level representation learning through graph aggregation. Built upon the enhanced sample-level features produced by ESP, PGC constructs a sample–prototype bipartite graph that integrates geometric neighborhood relations with nongeometric category priors. This design yields compact and robust category-aware representations, thereby improving discriminability in CDFSL HSIC. Specifically, the bipartite graph consists of two types of vertices: data samples, such as support and query features extracted by ESP, and prototype vectors representing category centroids in the metric space. By directly associating samples with their corresponding prototypes, the graph enables both geometric anchoring and nongeometric category expansion.

We construct a metric-guided bipartite graph in which prototypes serve as intermediaries to model global category structure. This design enables the graph topology to capture nongeometric information, including implicit correlations among samples linked to the same prototype and distributional commonalities of cross-domain samples in the prototype space, while incorporating inherent category priors of HSIs. Prototypes are computed as category centroids based on ESP-enhanced features. To simplify graph construction, PGC encodes sample–prototype relations into a category probability matrix $\mathbf{P}$, which is used as the bipartite edge weights. Consequently, the model jointly accounts for geometric distances to category centers and global interclass associations, forming the basis for category-aware representation learning. For labeled support samples, the corresponding rows of $\mathbf{P}$ are one-hot encoded, whereas for unlabeled query samples, distances to prototypes are converted into category probabilities via Softmax.

Considering the sparsity of the graph, we apply sparse processing to the probability matrix $\mathbf{P}$. Specifically, elements in the same row with probability values smaller than the threshold $1/n_{all}$, which corresponds to the expected uniform probability assuming equal class distribution, denote weak and unreliable associations. By constructing the graph, the topological relationships between these samples and other samples of the same category mediated by the prototype of the category are omitted, effectively suppressing noise while preserving meaningful structure and reducing graph complexity.

From the category probability matrix $\mathbf{P}$, we derive the bipartite adjacency matrix $\mathbf{G}$ to quantify sample-level associations mediated by prototype vectors. In the metric space, a sample's category probability is anchored to the prototype of that class and the ESP-enhanced features, with the independence of matrix elements ensuring clear separation of weights across categories. To integrate geometric neighborhood relations with nongeometric priors, the module applies the Markov chain rule: the edge weight between two samples x^m and x^n is determined by their joint associations with the same prototype C_l. Thus, $\mathbf{G}$ encodes sample–sample correlations through prototype-mediated connections

$$\mathbf{G}_{mn} = \mathbf{G}_{m\to n} = \sum_{i=1}^{n_l} \frac{p_{mi}p_{ni}}{p_{\cdot i}}. \tag{4}$$

Referring to this formula, the adjacency matrix of the graph $\mathbf{G}$ can be transformed into a symmetric form for easier network interpretation, which facilitates subsequent derivative calculations for gradient backpropagation and simplifies computational processes. In this context, $\mathbf{D}$ is a diagonal matrix to normalize the effect of prototype vectors. The sample–prototype bipartite graph integrates geometric and nongeometric priors to enhance sample compactness and global consistency for category-level feature extraction

$$\mathbf{G} = \mathbf{P}\mathbf{D}^{-1}\mathbf{P}. \tag{5}$$

To ensure the stability and reliability of category associations captured by the bipartite graph, a graph smoothness regularization is introduced to enforce similar category attributes for similar samples, while incorporating label information to optimize feature representations. The graph Laplacian matrix $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{G}\mathbf{D}^{-1/2}$ is then computed to quantify the smoothness S

$$S = \frac{1}{2}\sum_{i,j=1}^{n_{all}} p_{i,j}(y_i - y_j) = \mathbf{y}^T(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{y}^T\mathbf{L}\mathbf{y}. \tag{6}$$

Specifically, graph node features are updated via an iterative label propagation process to enforce category-level consistency. Within each episode, graph smoothness S is introduced as a regularization term, and the F-norm is used to quantify the discrepancy between predicted and true category distributions. The propagation terminates when the maximum change in sample-level label distributions between successive iterations falls below a predefined threshold, with an additional iteration limit imposed for stability. Based on the converged label distributions, the Laplacian matrix $\mathbf{L}$ is then updated in mini-batches. Finally, a cross-entropy loss between the propagated pseudo-labels and ground-truth labels is computed, allowing nongeometric relational information to constrain earlier network components and promote feature learning from the sample level to the category level

$$\mathbf{y} = \underset{\mathbf{y}}{argmin}\left(\left\|\mathbf{y} - \mathbf{y}^t\right\|_F^2 + \lambda S\right) \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

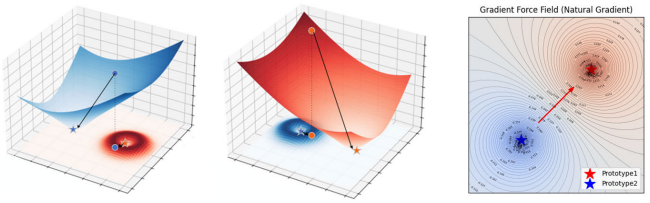


Fig. 2. Schematic of the gradient-aware metric function. (Left and middle) Prototype-induced attractive ($\partial \ell_{m^+}/\partial x_i$) (pull force) and repulsive effects ($\partial \ell_{m^-}/\partial x_i$) (push force) on hard positive and negative samples of the corresponding class under the guidance of the proposed gradient-aware metric function. The 3-D surfaces visualize the magnitude of the gradients, where larger values indicate stronger force-like influences—typically associated with distant intraclass positive samples or near-boundary interclass negative samples. (Right) Distribution of the force-field lines around each prototype vector, providing an intuitive characterization of the gradient-driven metric landscape.

where λ is the regularization parameter, and cross-entropy loss guided by nongeometric relationships is as follows:

$$\ell_{ng} = -\frac{1}{n_{all}} \sum_{i=1}^{n_{all}} y_i \log p(y_i). \tag{8}$$

Through the above regularization process, the model ultimately outputs features that not only preserve the fine-grained spectral information optimized by ESP but also integrate the category-level geometric and nongeometric associations captured by the bipartite graph, forming compact and robust category-level representations. These representations are capable of retaining the unique characteristics of individual samples while encoding the global graph topological priors, thereby significantly enhancing the discriminability and robustness of the extracted features and effectively improving classification reliability in CDFSL HSIC scenarios.

D. Gradient-Aware Metric Function

In the CDFSL HSIC task, the PGC module generates compact category-level features via a sample–prototype bipartite graph. Nevertheless, conventional scalar distance-based metrics remain a critical bottleneck, as they only quantify geometric discrepancies with a single value while ignoring directional information. This issue is particularly pronounced in cross-domain scenarios, where scalar distances cannot distinguish between a sample's tendency to approach the correct prototype and its need to move away from incorrect ones, leading to coarse sample–prototype modeling and reduced classification reliability.

To this end, we design a gradient-aware metric function that interprets gradients during parameter updates as directional vectors. By jointly considering their direction and magnitude, this function adaptively refines sample-prototype relations and addresses the lack of directional information in conventional scalar distances. The gradient direction reflects adjustment trends: for a positive sample of class i, low similarity to prototype C_i yields a gradient pointing toward C_i; for a negative sample, high similarity to C_i produces a gradient pointing away from it. The gradient magnitude further reflects adjustment strength, where larger values indicate stronger corrections, either pulling positive samples closer to their prototypes or pushing negative samples farther from incorrect ones. To initialize this process, the basic similarity between samples and prototypes is computed using the negative Euclidean distance, considering the high-dimensional nature of hyperspectral features, upon which the metric function ℓ_m is constructed. The detailed information is shown in Fig. 2

$$\ell_{m^+} = \frac{1}{|\mathbf{C}|} \sum_{i=1}^{N} log \left(1 + \sum_{j=1}^{n_{all}} e^{\alpha(\varphi - sim(x_j, C_i))}\right), \quad \text{if } x_j \in \mathbf{X}_i$$
$$\ell_{m^-} = \frac{1}{|\mathbf{C}|} \sum_{i=1}^{N} log \left(1 + \sum_{j=1}^{n_{all}} e^{\alpha(\varphi + sim(x_j, C_i))}\right), \quad \text{if } x_j \notin \mathbf{X}_i \tag{9}$$
$$\ell_m = \ell_{m^+} + \ell_{m^-} \tag{10}$$

where the gradient adjustment factor α controls the sensitivity of the loss to similarity variations, balancing the influence of hard and easy samples during optimization. A moderate value enhances informative samples near class boundaries while suppressing noise. The similarity threshold φ is empirically determined from the similarity distribution within each episode to distinguish ambiguous from confident sample–prototype relations. Together, these parameters stabilize training and improve the discriminative capability of the proposed metric. The similarity between a sample and a category is directly measured by the negative value of the Euclidean distance between them in the prototype network, that is, $sim(x_j, C_i)$.

Taking the derivatives of the two terms in the above equation yields the following results. It can be observed that for positive samples with low *sim* and negative ones with high *sim*, relatively large gradients emerge in ℓ_{m^+} and ℓ_{m^-}, respectively. By leveraging gradients to perceive sample movements in the metric space, positive samples are guided toward the vicinity of their corresponding category centers, while negative samples are driven to move away from the centers of irrelevant categories, thus adjusting the clustering distribution of the samples

$$\frac{\partial \ell_{m^+}}{\partial x_i} = \frac{1}{|\mathbf{C}|} \frac{\alpha \sum_{j=1}^{n_{all}} e^{\alpha(\varphi - sim(x_j, C_i))}}{1 + \sum_{j=1}^{n_{all}} e^{\alpha(\varphi - sim(x_j, C_i))}} * \left(-\frac{\partial sim(x_j, C_i)}{\partial x_i}\right)$$
$$\frac{\partial \ell_{m^-}}{\partial x_i} = \frac{1}{|\mathbf{C}|} \frac{\alpha \sum_{j=1}^{n_{all}} e^{\alpha(\varphi + sim(x_j, C_i))}}{1 + \sum_{j=1}^{n_{all}} e^{\alpha(\varphi + sim(x_j, C_i))}} * \left(\frac{\partial sim(x_j, C_i)}{\partial x_i}\right). \tag{11}$$

IV. Experiments

A. Experimental Dataset

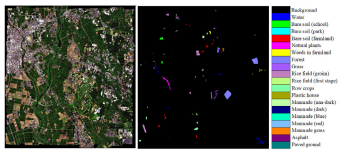
For the CDFSL HSIC task, the Chikusei dataset is selected as the SD, while the Pavia University (UP), Salinas (SA), Houston2013 (Hou13), and Indian Pines (IP) datasets serve as TDs.

1) SD Dataset: We selected the Chikusei dataset as the SD. The Chikusei dataset was acquired by the Hyperspec-VNIR-CIRIS spectrometer in Tsukuba, Ibaraki, Japan. The ground sampling distance is 2.5 m, and the scene consists of 2517 × 2335 pixels. The dataset contains 512 bands, with 128 bands in the spectral range of 363–1018 nm. There are a total of 19 land-cover categories. The detailed information of the Chikusei dataset can be found in Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I

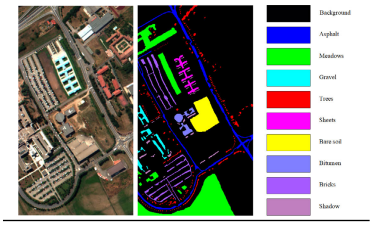
DETAILED SAMPLE COUNTS FOR EACH CLASS AND BRIEF PRESENTATION OF THE CHIKUSEI DATASETS



No.	Name	Count
1	Water	2345
2	Bare soil (school)	2859
3	Bare soil (park)	236
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (groom)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (nan-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

TABLE II

DETAILED SAMPLE COUNTS FOR EACH CLASS AND BRIEF PRESENTATION OF THE UP DATASETS



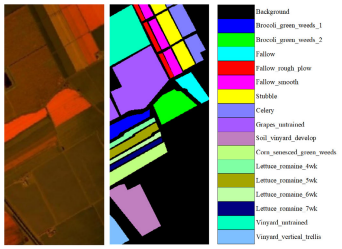
No.	Name	Count
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted Metal sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
	Total	42776

2) TD Dataset:

a) UP dataset: The UP dataset was obtained from the Reflected Optics System Imaging Spectrometer (ROSIS-3) sensor in Pavia, Italy, which has a size of 610 × 340 pixels and consists of 103 spectral bands. The content in the image is classified into nine categories, with a total of 42 776 labeled samples. The detailed number of samples for each category within the UP dataset, as well as the pseudo-color image and ground-truth map, are shown in Table II.

TABLE III

DETAILED SAMPLE COUNTS FOR EACH CLASS AND BRIEF PRESENTATION OF THE SALINAS (SA) DATASET



No.	Name	Count
1	Brocoli_green_ weeds_1	2009
2	Brocoli_green_ weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_ green_weeds	3728
11	Lettuce_Romaine_ 4wk	1068
12	Lettuce_Romaine_ 5wk	1927
13	Lettuce_Romaine_ 6wk	916
14	Lettuce_Romaine_ 7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_ trellis	1807
	Total	54129

b) Salinas (SA) dataset: The Salinas dataset was captured by the 224-band AVIRIS sensor over the Salinas Valley in California. It has a relatively high spatial resolution of 3.7 m. The covered area is 512 × 217, and the number of bands available in the dataset is 200. The ground truth contains 16 categories. The detailed number of samples for each category within the Salinas dataset, as well as the pseudo-color image and ground-truth map, are shown in Table III.

c) Houston2013 (Hou13) dataset: Houston2013 dataset was acquired from the campus of the University of Houston and its surrounding urban areas. It contains 144 spectral bands, with a wavelength range from 380 to 1050 nm. The spatial size is 349 × 1905, and there are a total of 15 land-cover categories. The detailed information within the Houston2013 dataset is shown in Table IV.

d) IP dataset: The IP dataset was acquired in 1992 by the airborne visible/infrared imaging spectrometer (AVIRIS) over an agricultural area in Northwestern Indiana, USA. The original data covers a spatial extent of 145 × 145 pixels. The number of usable spectral bands available in the dataset is 200. The dataset has a spatial resolution of 20 meters per pixel. In practical classification tasks, only 10 249 labeled pixels are utilized, representing 16 distinct land-cover classes. The detailed information of the IP dataset is presented in Table V.

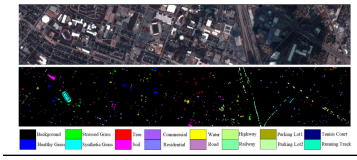
B. Experimental Setting

In terms of the selection of comparison algorithms, several state-of-the-art deep learning algorithms have been selected, including DCFSL [26], Gia-CFSL [28], CMFSL [27], DMCM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

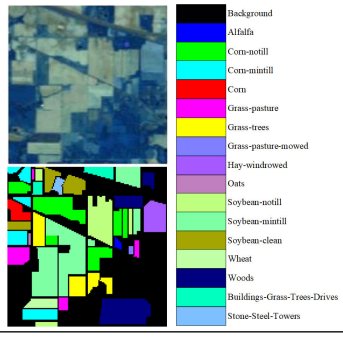
DETAILED SAMPLE COUNTS FOR EACH CLASS AND BRIEF PRESENTATION OF THE HOUSTON2013 (HOU13) DATASET



No.	Name	Count
1	Healthy Grass	1251
2	Stressed Grass	1254
3	Synthetic Grass	697
4	Tree	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot1	1233
13	Parking Lot2	469
14	Tennis Court	428
15	Running Track	660
	Total	15029

TABLE V

DETAILED SAMPLE COUNTS FOR EACH CLASS AND BRIEF PRESENTATION OF THE IP DATASET



No.	Name	Count
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	483
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	10249

[38] MLPA [39], RPCL [40], and FDFSL [41]. To ensure the fairness of the experimental results between the proposed algorithm and these comparative algorithms, in terms of specific experimental settings, the Chikusei dataset is used as

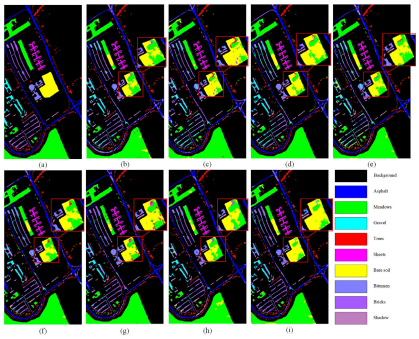


Fig. 3. Classification result on the UP dataset. (a) Ground truth. (b) DCFSL. (c) Gia-CFSL. (d) CMFSL. (e) MLPA. (f) DMCM. (g) RPCL. (h) FDFSL. (i) GRASP.

the SD dataset, and UP, Salinas (SA), IP, and Houston2013 (Hou13) are used as the TD datasets. During the training and testing process, the specific dataset settings are as follows: 200 labeled data are selected for each class of the SD dataset (for Chikusei, since only the number of samples in 18 categories exceeds 200, the final SD dataset is a dataset that contains a total of 18 classes with 3600 samples), when the TD dataset participates in FSL, five samples are selected from each class as the training dataset for FSL, and the data is augmented to 200 samples per class through data augmentation operations, while the remaining samples are used as the testing dataset for FSL episode. In the testing phase, the testing training set is input into the trained feature extractor, and the K-nearest neighbor classifier is used to classify the extracted features of the testing set. The network executes a total of 3000 episodes, using the Adam optimizer, and the learning rate is set to 0.001. In one episode training, a training task of N-way K-shot Q-query is constructed, where N is the total number of categories of the TD dataset ($N = 16$ for Salinas and IP, $N = 9$ for UP, and $N = 15$ for Hou13), K takes the value of 2, and Q takes the value of 19.

Regarding the selection of indicators for evaluating experimental results, we have chosen three commonly used indicators in HSIC, namely: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

C. Comparison of Experimental Results

In the CDFSL HSIC task, the proposed GRASP-Net demonstrates superior performance over existing methods across four benchmark datasets, validating its effectiveness and robustness. On the UP dataset containing nine categories, GRASP-Net achieves an OA of 86.31%, surpassing the second-best method FDFSL (84.60%) by a statistically significant margin of 1.71%, with a lower standard deviation (±2.56%), further confirming its robustness. This improvement benefits from the ESP module, which performs dynamic spectral reorganization through the kernel-attentive and kernel-scattered dual-branch design, enabling the model to capture subtle yet discriminative spectral variations at the sample level. The separability of classes such as trees (94.77%) and painted metal sheets (99.34%) is significantly improved. Detailed results are shown in Fig. 3 and Table VI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

CLASSIFICATION PERFORMANCE (%) ON THE UP DATASET. THE OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH RED/PINK AND BLUE/CYAN, RESPECTIVELY

No.	DCFSL	Gia-CFSL	CMFSL	MLPA	DMCM	RPCL	FDFSL	GRASP
1	75.67±15.88	78.13±8.34	77.59±11.76	80.74±11.22	80.61±11.81	83.06±5.42	85.99±8.34	85.36±5.50
2	86.03±7.56	87.90±6.97	85.82±6.32	88.27±6.34	86.11±7.35	86.13±7.38	82.83±9.63	88.58±5.86
3	57.80±13.75	54.24±9.88	64.27±11.50	66.99±16.94	63.43±15.51	69.83±9.54	67.25±7.95	66.99±11.65
4	93.31±1.89	89.43±5.92	92.34±2.36	91.44±2.69	92.81±2.96	90.12±8.02	92.02±2.19	94.77±2.23
5	98.88±0.81	98.34±1.83	98.82±1.04	98.49±1.20	98.71±1.35	98.95±1.19	99.08±0.24	99.34±0.83
6	67.86±13.99	76.05±11.53	81.82±7.63	73.99±8.32	76.20±12.56	75.41±12.05	67.31±14.19	78.71±13.02
7	79.00±7.77	76.54±8.01	87.82±7.07	86.55±7.40	92.30±7.12	88.98±10.23	91.65±4.22	85.11±7.23
8	61.62±16.00	68.38±13.89	67.55±12.97	67.72±15.98	76.77±7.54	88.11±4.73	84.28±4.73	84.60±3.24
9	98.82±2.34	96.75±6.35	99.69±0.50	99.09±1.92	98.17±3.28	99.25±1.11	96.42±2.62	97.54±4.74
OA	78.79±3.33	81.94±2.33	82.68±2.37	82.97±2.54	83.51±2.73	83.73±3.80	84.60±3.61	86.31±2.56
AA	79.89±2.12	80.64±1.75	83.97±1.94	83.33±2.49	85.01±2.51	86.65±2.68	85.20±1.57	86.78±2.37
Kappa	73.71±3.92	76.46±2.69	77.57±2.49	75.16±3.07	78.57±3.23	80.30±2.68	81.31±3.22	82.10±1.86

TABLE VII

CLASSIFICATION PERFORMANCE (%) ON THE SA DATASET. THE OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH RED/PINK AND BLUE/CYAN, RESPECTIVELY

No.	DCFSL	CMFSL	Gia-CFSL	MLPA	DMCM	RPCL	FDFSL	GRASP
1	98.43±1.72	97.34±2.82	99.35±0.79	98.11 ±1.74	98.25±0.69	99.58±0.60	99.36±0.83	99.08±0.74
2	99.72±0.21	99.81±0.35	99.36±0.44	99.45 ±0.42	97.95±3.93	99.86±0.25	98.13±5.50	99.90±0.12
3	90.46±8.66	94.91±5.06	91.32±6.10	90.41 ±8.90	91.10±6.88	93.08±8.64	84.53±12.03	91.95±7.36
4	99.30±0.82	99.25±0.99	98.94±1.12	99.27 ±0.86	99.03±0.34	99.73±0.33	99.34±0.60	99.04±1.76
5	91.40±3.46	92.63±3.15	88.06±6.14	90.55 ±3.57	93.64±3.42	89.47±5.30	94.41±3.21	90.07±5.26
6	98.94±0.95	98.90±1.22	97.93±1.50	99.50 ±0.45	98.44±0.50	98.41±1.35	99.12±0.56	96.45±2.34
7	99.17±1.39	98.97±1.63	98.04±4.16	99.67 ±0.45	99.78±0.30	99.47±0.76	99.33±1.35	99.27±1.50
8	66.43±14.61	69.39±16.76	72.83±15.78	73.18 ±8.59	74.76±6.56	75.77±10.94	77.31±7.16	78.99±8.13
9	99.61±0.36	99.35±0.75	99.18±0.74	99.41 ±0.58	99.39±0.86	99.81±0.52	99.37±1.34	99.83±0.30
10	73.77±14.44	73.60±14.89	73.91±11.06	69.99 ±16.96	84.37±8.77	84.45±6.81	83.58±8.56	86.40±6.06
11	95.36±6.05	95.32±4.10	92.62±6.97	95.83 ±3.62	96.31±4.98	97.73±4.43	97.28±1.55	98.89±0.88
12	99.28±1.16	99.07±1.82	98.77±1.13	97.47 ±3.26	99.30±0.96	99.46±0.50	98.67±1.78	99.75±0.28
13	98.36±1.33	98.52±1.88	98.77±1.13	98.58 ±0.96	98.82±1.27	98.56±2.68	98.44±1.37	98.89±0.92
14	98.34±1.90	99.48±0.38	97.54±2.97	98.34 ±2.00	99.33±1.40	97.97±2.64	99.55±0.36	98.27±2.30
15	76.30±7.05	78.49±3.80	77.83±6.66	74.64 ±4.28	81.18±8.26	81.65±1.75	81.31±6.70	81.56±4.21
16	88.84±7.16	89.22±7.77	88.11±7.16	90.11±5.46	92.24±4.16	90.40±7.11	91.08±5.62	90.50±4.95
OA	86.63±2.60	87.70±2.01	87.72±2.88	88.56 ±2.64	89.12±1.31	90.12±1.94	90.26±1.72	91.24±1.56
AA	92.10±1.23	92.77±1.14	91.96±1.13	92.16 ±1.53	93.76±0.92	94.09±0.85	93.80±1.31	94.30±0.87
Kappa	85.18±2.85	86.37±3.27	86.38±3.23	87.18 ±2.91	89.03±1.45	89.02±2.13	89.05±1.90	89.71±1.72

TABLE VIII

CLASSIFICATION PERFORMANCE (%) ON THE HOU13 DATASET. THE OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH RED/PINK AND BLUE/CYAN, RESPECTIVELY

No.	DCFSL	CMFSL	Gia-CFSL	MLPA	DMCM	RPCL	FDFSL	GRASP
1	86.69±4.93	86.20 ± 4.28	85.06±4.23	87.38±5.89	85.91 ±5.67	83.03±8.65	87.06±6.11	87.99±7.39
2	75.36±10.62	83.85 ± 8.14	84.27±7.27	80.97±6.76	88.08 ±5.23	88.09±5.46	84.72±9.11	88.69±5.71
3	97.24±1.83	95.62 ± 2.21	97.30±3.04	96.24±1.81	95.90 ±2.15	97.59±2.00	98.16±1.17	96.40±2.69
4	91.21±4.37	91.65 ± 4.58	91.52±3.71	92.44±2.62	91.35 ±2.82	93.19±2.66	92.91±2.48	92.05±4.24
5	96.55±4.61	96.56 ± 0.47	95.18±2.91	96.65±3.08	97.49 ±0.60	98.85±1.53	98.67±0.75	98.72±0.97
6	77.38±10.83	80.34 ± 12.81	75.06±13.90	78.91±11.76	86.28 ±4.13	87.50±5.73	83.81±2.91	89.31±6.55
7	62.98±9.69	66.86 ± 8.69	57.51±4.55	65.67±11.32	75.03 ±12.52	79.05±10.35	73.79±5.26	82.13±5.11
8	41.54±7.26	42.69 ± 6.39	39.91±5.75	44.73±5.60	47.48 ±7.24	44.47±8.06	46.99±5.87	40.32±7.35
9	60.58±6.31	65.43 ± 5.87	64.60±9.44	55.75±9.23	72.51 ±6.59	76.88±9.81	72.70±7.52	76.64±11.80
10	58.99±13.78	63.11 ± 12.02	61.56±13.49	61.55±13.13	65.94 ±17.93	60.65±16.58	64.92±15.99	67.17±16.44
11	58.28±7.23	59.47 ± 7.48	58.33±11.42	53.63±9.22	53.94 ±11.86	62.59±11.58	67.19±8.55	68.31±7.69
12	64.76±11.34	59.34 ± 11.30	62.81±10.31	68.59±14.49	64.06 ±10.01	60.90±11.27	67.71±15.84	64.97±15.84
13	83.08±8.27	86.29 ± 3.79	88.62±7.54	78.90±14.73	90.65 ±2.13	96.81±2.29	94.53±2.27	97.13±1.72
14	84.16±8.66	90.55 ± 6.77	85.18±8.47	83.26±5.27	91.81 ±4.99	88.13±10.08	94.36±4.93	91.87±7.03
15	90.08±6.20	90.39 ± 5.76	89.39±7.61	91.94±7.09	95.04 ±3.98	95.24±3.65	98.88±0.56	98.05±1.91
OA	72.87±1.97	73.23 ± 2.19	73.30±2.00	73.65±1.81	77.32 ±2.33	78.01±1.98	79.04±2.12	79.92±1.95
AA	75.26±1.77	75.99 ± 2.08	75.75±1.08	75.77±1.85	80.03 ±2.11	80.86±2.02	81.76±1.69	82.65±1.84
Kappa	70.70±2.11	75.97 ± 2.34	71.19±2.15	71.53±1.96	75.65 ±2.51	76.37±2.13	77.41±2.28	78.34±2.10

In Table VII, GRASP-Net sets a new benchmark with an OA of 91.24% on the Salinas (SA) dataset with 16 agricultural categories, outperforming the second-best method by 1.0%. Attributed to the PGC module, it incorporates global graph

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

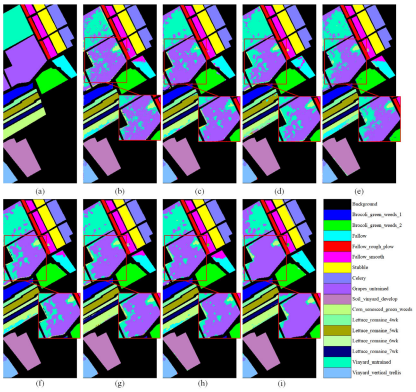


Fig. 4. Classification result on the SA dataset. (a) Ground truth. (b) DCFSL. (c) Gia-CFSL. (d) CMFSL. (e) MLPA. (f) DMCM. (g) RPCL. (h) FDFSL. (i) GRASP.

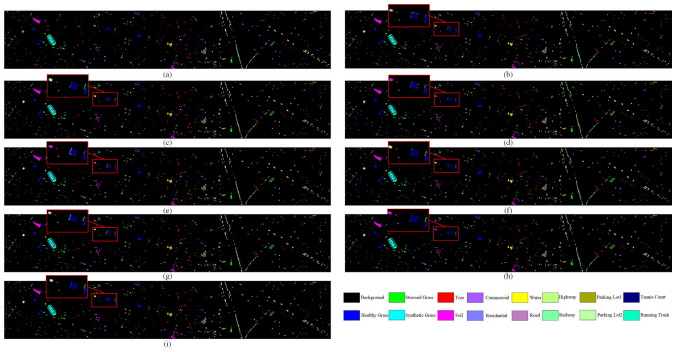


Fig. 5. Classification result on the HOU13 dataset. (a) Ground truth. (b) DCFSL. (c) Gia-CFSL. (d) CMFSL. (e) MLPA. (f) DMCM. (g) RPCL. (h) FDFSL. (i) GRASP.

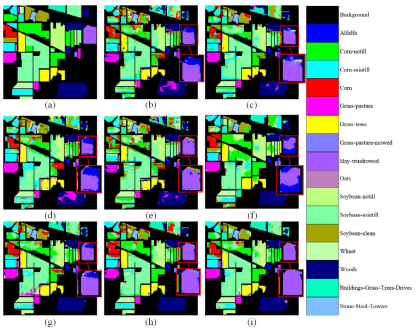


Fig. 6. Classification result on the IP dataset. (a) Ground truth. (b) DCFSL. (c) Gia-CFSL. (d) CMFSL. (e) MLPA. (f) DMCM. (g) RPCL. (h) FDFSL. (i) GRASP.

topology and category-level priors to refine sample features into more consistent category representations. As a result, the Vineyard untrained class achieves an accuracy of 81.56% (5.26% higher than DCFSL), while small-sample categories such as Celery (99.27%) achieve nearly perfect recognition. The visual results in Fig. 4 further show fewer misclassifications in vegetation-mixed regions compared with other approaches.

On the HOU13 dataset, which consists of 15 categories with significant spectral heterogeneity, GRASP-Net achieves an OA of 79.92% and a Kappa coefficient of 78.34% in Table VIII, outperforming FDFSL (79.04%) by 0.88%, thereby demonstrating strong robustness to domain shift. By introducing the gradient-aware metric function, the metric space is further regularized with gradient-induced class-force information, enabling more reliable sample–prototype associations and sharper decision boundaries. Consequently, the accuracies of challenging classes such as Healthy Grass (87.99%) and Water (89.31%) are improved by 1.3% and 12% over DCFSL, respectively, while misclassifications among spectrally similar categories such as Highway and Railway are effectively reduced. Classification results are shown in Fig. 5.

As shown in Table IX, GRASP-Net exhibits remarkable generalization capability on the IP dataset with 16 categories under extremely limited sample conditions, achieving an OA of 74.06%. Notably, the method reaches 100% accuracy on the Oats class (20 samples) and 89.94% on the Hay-windrowed class (478 samples). Furthermore, the proposed hierarchical feature extraction pipeline and boundary-aware decision structure significantly enhance the classification of categories such as Woods (91.43%). The visualization results in Fig. 6 show that GRASP-Net produces predictions with coherent, well-defined region boundaries, with no noticeable scattered isolated pixel artifacts, which are often present in baseline methods.

In summary, compared with DCFSL, CMFSL, Gia-CFSL, and other methods, GRASP-Net explicitly models category-level relations via a prototype-guided graph and refines them with gradient-aware metric learning, yielding more stable and discriminative representations under domain shift. Its superior performance stems from the complementary synergy of three core modules: the ESP module enhances sample-level spectral discriminability, while the PGC module incorporates global graph topology priors beyond geometric neighborhoods to strengthen category-level representations. Furthermore, the gradient-aware metric function adjusts sample–prototype relations through gradient-induced forces, improving intraclass compactness and interclass separability. Together, these mechanisms enable robust CDFSL HSIC classification under small sample and high-heterogeneity conditions.

D. Ablation Study

To evaluate the effectiveness of the three core modules in GRASP-Net, V1: ESP module, V2: PGC, and V3: gradient-aware metric function. We conducted ablation experiments on four benchmark datasets. Each component was removed in turn to analyze its contribution, and the results are summarized in Table X.

On the UP dataset, using only V1 improves the OA from 84.32% to 84.99% and slightly increases the AA from 85.47% to 85.66%, indicating that the ESP module alone provides limited gains by enhancing sample-level spectral representations. Adding V2 on top of V1 further raises the OA to 85.37% and the AA to 87.14%, revealing a clear synergistic effect between ESP and PGC through joint modeling of sample-level spectral features and category-level structural information. On the SA dataset, combining V1 and V2 increases the OA from 89.35% to 90.49% and the Kappa from 88.27% to 89.19%, showing that PGC effectively extracts category-level

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CLASSIFICATION PERFORMANCE (%) ON THE IP DATASET. THE OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH RED/PINK AND BLUE/CYAN, RESPECTIVELY

No.	DCFSL	CMFSL	Gia-CFSL	MLPA	DMCM	RPCL	FDFSL	GRASP
1	95.12±5.12	97.07±3.24	93.90±7.81	96.22±4.13	97.32±4.42	99.51±0.98	99.76±0.73	99.76±0.73
2	39.79±6.82	44.17±7.59	44.17±10.75	46.20±8.23	59.22±12.50	57.60±10.19	64.77±3.57	62.90±12.08
3	49.44±11.33	52.69±9.53	49.42±8.71	56.96±13.10	63.25±10.47	59.72±7.58	65.36±1.80	65.50±7.98
4	74.05±14.68	74.70±14.74	75.82±16.70	73.66±11.60	84.48±9.00	84.31±9.37	88.53±4.83	89.96±9.55
5	72.70±7.29	80.61±7.38	76.09±8.34	65.36±13.09	89.16±5.42	80.50±6.57	72.64±0.78	83.97±9.45
6	79.16±6.86	74.98±8.43	79.38±7.90	77.17±6.57	80.69±7.56	86.86±7.24	92.88±2.76	86.03±6.92
7	98.70±2.78	98.70±2.78	98.70±1.99	92.61±7.79	99.13±2.61	98.70±2.78	100.00±0.00	100.00±0.00
8	70.87±9.49	71.73±15.20	73.19±11.21	68.56±13.38	86.11±12.35	89.64±6.48	58.33±2.78	89.94±8.32
9	100.00±0.00	99.23±2.10	99.33±2.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
10	54.75±15.35	57.67±8.44	56.79±12.08	58.00±8.76	65.63±6.64	65.63±5.75	69.79±5.05	69.89±6.34
11	58.93±8.49	58.98±7.80	64.10±9.83	62.94±6.02	59.16±6.95	65.20±9.42	54.65±3.07	68.35±10.19
12	39.49±6.29	54.06±7.41	50.39±8.10	39.10±6.19	64.05±11.23	48.76±13.56	57.35±3.68	59.56±10.95
13	96.40±6.97	98.35±3.49	98.60±2.12	97.35±2.60	97.35±3.61	96.80±3.76	99.45±0.57	97.00±3.03
14	87.89±8.22	83.04±9.99	82.90±10.15	85.33±9.73	84.38±10.56	90.53±7.74	90.48±2.01	91.43±6.63
15	59.45±10.11	70.73±12.30	68.19±14.62	75.20±9.56	73.18±12.75	81.42±9.40	90.89±4.12	85.49±7.76
16	99.55±0.56	99.66±0.52	99.43±1.37	98.30±2.84	98.86±2.03	98.30±3.02	99.77±0.45	98.64±2.02
OA	61.94±2.70	63.95±3.32	64.69±3.32	70.21±1.28	70.43±2.69	71.54±2.70	72.07±0.93	74.06±2.66
AA	73.52±1.73	76.02±1.82	75.65±1.76	77.03±1.39	81.37±1.82	81.47±1.40	80.79±0.64	84.27±1.32
Kappa	57.14±2.91	59.40±3.54	60.18±3.60	65.05±1.37	66.84±2.94	67.91±2.89	68.77±1.00	71.89±2.85

TABLE X
CLASSIFICATION PERFORMANCE (%) OF THE ABLATIONS ON THE PROPOSED METHOD

		OA	AA	Kappa
UP	base	84.32±4.23	85.47±3.64	79.70±5.18
	v1	84.99±2.01	85.66±3.17	80.41±2.49
	v1+v2	85.37±2.47	87.14±2.32	81.05±2.86
	v1+v2 +v3	86.31±2.56	86.78±2.37	82.10±1.86
SA	base	89.35±1.98	93.61±1.12	88.27±1.75
	v1	89.72±1.98	93.75±0.97	88.83±2.21
	v1+v2	90.49±2.47	94.13±0.92	89.19±2.01
	v1+v2+v3	91.24±1.56	94.30±0.87	89.71±1.72
HOU13	base	78.12±1.75	80.91±1.97	76.89±1.85
	v1	78.76±1.45	81.55±1.89	77.21±2.32
	v1+v2	79.23±1.89	81.63±1.62	78.93±1.76
	v1+v2+v3	79.92±1.95	82.65±1.84	78.34±2.10
IP	base	71.03±2.32	81.35±1.13	67.41±2.45
	v1	71.62±2.50	81.68±1.37	67.93±3.15
	v1+v2	73.83±1.67	83.94±1.04	71.62±1.79
	v1+v2+v3	74.06±2.66	84.27±1.32	71.89±2.85

features by incorporating nongeometric metric information and improves robustness under domain shifts. With V3 included, the complete model (V1 + V2 + V3) achieves the best performance across all datasets. Specifically, OA on the UP dataset reaches 86.31%, while on HOU13, V3 refines sample–prototype relations via gradient-driven optimization, achieving 79.92% OA and 82.65% AA. On the IP dataset, the Kappa improves from 71.62% to 71.89%, confirming the effectiveness of V3 in small-sample settings.

V. CONCLUSION

This article proposes a hierarchical framework for CDFSL HSIC. It enhances spectral features to preserve discriminative information while reducing redundancy and introduces a prototype-guided graph to integrate local geometry with category-level priors. Moreover, a gradient-aware metric function is designed to refine sample–prototype relations by modeling both adjustment direction and magnitude. Together, these components yield representations that are fine-grained at the sample level and discriminative at the category level. In future work, we will extend the framework toward spatiotemporal hyperspectral data and investigate its application to large-scale and dynamic remote sensing tasks.

REFERENCES

[1] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[2] S. Veraverbeke et al., "Hyperspectral remote sensing of fire: State-of-the-art and future perspectives," *Remote Sens. Environ.*, vol. 216, pp. 105–121, Oct. 2018.

[3] R. Fakhrullin, L. Nigamatzyanova, and G. Fakhrullina, "Dark-field/hyperspectral microscopy for detecting nanoscale particles in environmental nanotoxicology research," *Sci. Total Environ.*, vol. 772, Jun. 2021, Art. no. 145478.

[4] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.

[5] K. Huang, S. Li, X. Kang, and L. Fang, "Spectral–spatial hyperspectral image classification based on KNN," *Sens. Imag.*, vol. 17, no. 1, pp. 1–13, Dec. 2016.

[6] O. Okwuashi and C. E. Ndehedehe, "Deep support vector machine for hyperspectral image classification," *Pattern Recognit.*, vol. 103, Jul. 2020, Art. no. 107298.

[7] J. Xia, P. Du, X. He, and J. Chanussot, "Hyperspectral remote sensing image classification based on rotation forest," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 239–243, Jan. 2014.

[8] Y. Zhang, G. Cao, X. Li, and B. Wang, "Cascaded random forest for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1082–1094, Apr. 2018.

[9] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

[10] N. Falco, J. A. Benediktsson, and L. Bruzzone, "Spectral and spatial classification of hyperspectral images based on ICA and reduced morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6223–6240, Nov. 2015.

[11] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.

[12] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[13] M. Xu et al., "Multisource adaptive feature enhancement and fusion network for coastal wetland classification with hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 27798–27813, 2025.

[14] J. Ma, B. Li, H. Li, S. Meng, R. Lu, and S. Mei, "Remote sensing change detection by pyramid sequential processing with Mamba," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 19481–19495, 2025.

[15] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, May 2021.

[16] S. Ravi and H. Larochelle, "Optimization as a model for few-shot learning," in *Proc. Int. Conf. Learn. Represent.*, Toulon, France, Apr. 2017.

[17] W. Jiang, K. Huang, J. Geng, and X. Deng, "Multi-scale metric learning for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1091–1102, Mar. 2021.

[18] H. Tang, Y. Li, X. Han, Q. Huang, and W. Xie, "A spatial–spectral prototypical network for hyperspectral remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 167–171, Jan. 2020.

[19] J. Zeng, Z. Xue, L. Zhang, Q. Lan, and M. Zhang, "Multistage relation network with dual-metric for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510017.

[20] J. Wang and Y. Zhai, "Prototypical Siamese networks for few-shot learning," in *Proc. IEEE 10th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*, Jul. 2020, pp. 178–181.

[21] A. Qin et al., "Few-shot learning with prototype rectification for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409015.

[22] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[23] Z. Xue, Y. Zhou, and P. Du, "S^3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[24] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502418.

[25] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2960–2967.

[26] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[27] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[28] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.

[29] C. Ding et al., "GLGAT-CFSL: Global-local graph attention network-based cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522519.

[30] Z. Feng, S. Tong, S. Yang, X. Zhang, and L. Jiao, "Pseudo-label-assisted subdomain adaptation for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4729–4744, Jun. 2024.

[31] H. Xu et al., "Enhancing information maximization with distance-aware contrastive learning for source-free cross-domain few-shot learning," *IEEE Trans. Image Process.*, vol. 33, pp. 2058–2073, 2024.

[32] M. Gong et al., "RAFNet: Interdomain representation alignment and fine-tuning for image series classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4704416.

[33] X. Liu et al., "Deep unsupervised domain adaptation: A review of recent advances and perspectives," *APSIPA Trans. Signal Inf. Process.*, vol. 11, no. 1, pp. 1–51, Aug. 2022.

[34] J. Zheng et al., "A two-stage adaptation network (TSAN) for remote sensing scene classification in single-source-mixed-multiple-target domain adaptation (S^2M^2T DA) scenarios," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609213.

[35] M. Ghifary, W. B. Kleijn, M. Zhang, D. Balduzzi, and W. Li, "Deep reconstruction-classification networks for unsupervised domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 597–613.

[36] J. Wang and Y. Zhai, "Prototypical Siamese networks for few-shot learning," in *Proc. IEEE 10th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*, Jul. 2020, pp. 178–181.

[37] Z. Yang et al., "Label propagation and contrastive regularization for semisupervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609818.

[38] L. Hu, W. He, L. Zhang, and H. Zhang, "Cross-domain meta-learning under dual-adjustment mode for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526416.

[39] H. Liu, J. He, Y. Li, and Y. Bi, "Multilevel prototype alignment for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4400115.

[40] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[41] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:13:37 UTC from IEEE Xplore. Restrictions apply.